CLASS 9 MATHEMATICS

GANITA MANJARI

CHAPTER 4 – EXPLORING ALGEBRAIC IDENTITIES

END-OF-CHAPTER EXERCISES

1. Use suitable identities to find the following products:

(i) $(-3x+4)^2$

(ii) $(2s+7)(2s-7)$

(iii) $\left(p^2+\frac{1}{2}\right)\left(p^2-\frac{1}{2}\right)$

(iv) $(2n+7)(2n-7)$

(v) $(s-2t)(s^2+2st+4t^2)$

(vi) $\left(\frac{1}{2r}-4r\right)^2$

(vii) $(-3m+4k-l)^2$

(viii) $\left(x-\frac{1}{3}y\right)^3$

(ix) $\left(\frac{7}{2}k-\frac{2}{3}m\right)^3$

Solution:

(i)

$$(-3x+4)^2 = 9x^2 - 24x + 16.$$

(ii) Using $(a+b)(a-b) = a^2 - b^2$,

$$(2s+7)(2s-7) = (2s)^2 - 7^2 = 4s^2 - 49.$$

(iii)

$$\left(p^2 + \frac{1}{2}\right)\left(p^2 - \frac{1}{2}\right) = (p^2)^2 - \left(\frac{1}{2}\right)^2 = p^4 - \frac{1}{4}.$$

(iv)

$$(2n+7)(2n-7) = (2n)^2 - 7^2 = 4n^2 - 49.$$

(v) Using

$$(a-b)(a^2+ab+b^2) = a^3 - b^3,$$

we get

$$(s-2t)(s^2+2st+4t^2) = s^3 - (2t)^3 = s^3 - 8t^3.$$

(vi) Using $(a-b)^2 = a^2 - 2ab + b^2$,

$$\left(\frac{1}{2r} - 4r\right)^2 = \frac{1}{4r^2} - 4 + 16r^2.$$

(vii) Using

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca,$$

$$(-3m+4k-l)^2 = 9m^2 + 16k^2 + l^2 - 24mk - 8kl + 6ml.$$

(viii) Using $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$,

$$\left(x - \frac{1}{3}y\right)^3 = x^3 - x^2y + \frac{1}{3}xy^2 - \frac{y^3}{27}.$$

(ix)

$$\left(\frac{7}{2}k - \frac{2}{3}m\right)^3 = \frac{343}{8}k^3 - \frac{49}{2}k^2m + \frac{14}{3}km^2 - \frac{8}{27}m^3.$$

2. Find the values using suitable identities:

(i) 17×21

(ii) 104×96

(iii) 24×16

(iv) 147^3

(v) 199^3

(vi) 127^3

(vii) $(-107)^3$

(viii) $(-299)^3$

Solution:

(i)

$$17 \times 21 = (19-2)(19+2) = 19^2 - 2^2 = 361 - 4 = \boxed{357}.$$

(ii)

$$104 \times 96 = (100+4)(100-4) = 100^2 - 4^2 = 10000 - 16 = \boxed{9984}.$$

(iii)

$$24 \times 16 = (20+4)(20-4) = 20^2 - 4^2 = 400 - 16 = \boxed{384}.$$

(iv)

$$\begin{aligned} 147^3 &= (150-3)^3 \\ &= 150^3 - 3(150)^2(3) + 3(150)(3^2) - 3^3 \\ &= \boxed{3176523}. \end{aligned}$$

(v)

$$\begin{aligned} 199^3 &= (200-1)^3 \\ &= 200^3 - 3(200)^2 + 3(200) - 1 \\ &= \boxed{7880599}. \end{aligned}$$

(vi)

$$\begin{aligned} 127^3 &= (130-3)^3 \\ &= 130^3 - 3(130)^2(3) + 3(130)(3^2) - 3^3 \\ &= \boxed{2048383}. \end{aligned}$$

(vii)

$$(-107)^3 = -(107)^3.$$

Now,

$$107^3 = (100+7)^3 = 1225043.$$

Hence,

$$\boxed{(-107)^3 = -1225043}.$$

(viii)

$$(-299)^3 = -(299)^3.$$

Now,

$$299^3 = (300-1)^3 = 26730899.$$

Therefore,

$$\boxed{(-299)^3 = -26730899}.$$

3. Factor the following algebraic expressions:

(i) $4y^2 + 1 + \frac{1}{16y^2}$

(ii) $9m^2 - \frac{1}{25n^2}$

(iii) $27b^3 - \frac{1}{64b^3}$

(iv) $x^2 + \frac{5}{6}x + \frac{1}{6}$

(v) $27u^3 - \frac{1}{125} - \frac{27u^2}{5} + \frac{9u}{25}$

(vi) $64y^3 + \frac{1}{125}z^3$

(vii) $p^3 + 27q^3 + r^3 - 9pqr$

(viii) $9m^2 - 12m + 4$

(ix) $9x^3 - \frac{8}{3}y^3 + \frac{1}{3}z^3 + 6xyz$

(x) $4x^2 + 9y^2 + 36z^2 + 12xz + 36yz + 24xy$

(xi) $27u^3 - \frac{1}{216} - \frac{9}{2}u^2 + \frac{1}{4}u$

Solution:

(i)

$$4y^2 + 1 + \frac{1}{16y^2} = \left(2y + \frac{1}{4y}\right)^2.$$

(ii)

$$9m^2 - \frac{1}{25n^2} = \left(3m + \frac{1}{5n}\right)\left(3m - \frac{1}{5n}\right).$$

(iii)

$$27b^3 - \frac{1}{64b^3} = \left(3b - \frac{1}{4b}\right)\left(9b^2 + \frac{3}{4} + \frac{1}{16b^2}\right).$$

(iv)

$$\begin{aligned} x^2 + \frac{5}{6}x + \frac{1}{6} &= x^2 + \frac{1}{2}x + \frac{1}{3}x + \frac{1}{6} \\ &= x\left(x + \frac{1}{2}\right) + \frac{1}{3}\left(x + \frac{1}{2}\right) \\ &= \boxed{\left(x + \frac{1}{3}\right)\left(x + \frac{1}{2}\right)}. \end{aligned}$$

(v)

$$27u^3 - \frac{27u^2}{5} + \frac{9u}{25} - \frac{1}{125}$$

$$= (3u)^3 - 3(3u)^2\left(\frac{1}{5}\right) + 3(3u)\left(\frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^3$$

$$= \boxed{\left(3u - \frac{1}{5}\right)^3}.$$

(vi)

$$\begin{aligned} 64y^3 + \frac{z^3}{125} &= (4y)^3 + \left(\frac{z}{5}\right)^3 \\ &= \left(4y + \frac{z}{5}\right)\left(16y^2 - \frac{4}{5}yz + \frac{z^2}{25}\right). \end{aligned}$$

(vii)

$$p^3 + 27q^3 + r^3 - 9pqr$$

$$= p^3 + (3q)^3 + r^3 - 3(p)(3q)(r)$$

$$= \boxed{(p + 3q + r)\left(p^2 + 9q^2 + r^2 - 3pq - 3qr - pr\right)}.$$

(viii)

$$\begin{aligned} 9m^2 - 12m + 4 &= (3m)^2 - 2(3m)(2) + 2^2 \\ &= \boxed{(3m - 2)^2}. \end{aligned}$$

(ix)

$$9x^3 - \frac{8}{3}y^3 + \frac{1}{3}z^3 + 6xyz$$

$$= \frac{1}{3}\left(27x^3 - 8y^3 + z^3 + 18xyz\right)$$

$$= \frac{1}{3}\left[(3x)^3 + (-2y)^3 + z^3 - 3(3x)(-2y)z\right]$$

4. Simplify the following:

(i) $\dfrac{4x^2 + 4x + 1}{4x^2 - 1}$

(ii) $\dfrac{9(3a^3 - 24b^3)}{9a^2 - 36b^2}$

(iii) $\dfrac{s^3 + 125t^3}{s^2 - 2st - 35t^2}$

Solution:

(i)

$$\frac{4x^2 + 4x + 1}{4x^2 - 1} = \frac{(2x+1)^2}{(2x+1)(2x-1)}$$
$$= \boxed{\frac{2x+1}{2x-1}}.$$

(ii)

$$\frac{9(3a^3 - 24b^3)}{9a^2 - 36b^2} = \frac{3(a^3 - 8b^3)}{a^2 - 4b^2}$$
$$= \frac{3(a-2b)(a^2 + 2ab + 4b^2)}{(a-2b)(a+2b)}$$
$$= \boxed{\frac{3(a^2 + 2ab + 4b^2)}{a + 2b}}.$$

(iii)

$$\frac{s^3 + 125t^3}{s^2 - 2st - 35t^2}$$
$$= \frac{(s+5t)(s^2 - 5st + 25t^2)}{(s+5t)(s-7t)}$$
$$= \boxed{\frac{s^2 - 5st + 25t^2}{s - 7t}}.$$

5. Find possible expressions for the length and breadth of each rectangle whose area is given by:

(i) $25a^2 - 30ab + 9b^2$

(ii) $36s^2 - 49t^2$

Solution:

(i)

$$25a^2 - 30ab + 9b^2 = (5a)^2 - 2(5a)(3b) + (3b)^2$$
$$= \boxed{(5a - 3b)^2}.$$

Hence, possible length and breadth are

$$\boxed{5a - 3b, \quad 5a - 3b}.$$

(ii)

$$36s^2 - 49t^2 = (6s)^2 - (7t)^2$$
$$= \boxed{(6s + 7t)(6s - 7t)}.$$

Hence, possible length and breadth are

$$\boxed{6s + 7t, \quad 6s - 7t}.$$

6. Find possible expressions for the length, breadth and height of each cuboid whose volume is given by:

(i) $6a^2 - 24b^2$

(ii) $3ps^2 - 15ps + 12p$

Solution:

(i)

$$6a^2 - 24b^2 = 6(a^2 - 4b^2)$$
$$= 6(a + 2b)(a - 2b).$$

Hence, possible dimensions are

$$\boxed{6, \quad a + 2b, \quad a - 2b}.$$

(ii)

$$3ps^2 - 15ps + 12p = 3p(s^2 - 5s + 4)$$
$$= 3p(s - 1)(s - 4).$$

Hence, possible dimensions are

$$\boxed{3p, \quad s - 1, \quad s - 4}.$$

7. The village playground is shaped as a square of side 40 metres. A path of width s metres is created around the playground for people to walk. Find an expression for the area of the path in terms of s.

Solution:
Side of playground:

$$40 \text{ m}.$$

Since the path has width s metres on all four sides, the side of the outer square is

$$40 + 2s.$$

Area of outer square:

$$(40 + 2s)^2.$$

Area of playground:

$$40^2 = 1600.$$

Therefore,

$$\text{Area of path} = (40 + 2s)^2 - 1600$$

$$= 1600 + 160s + 4s^2 - 1600$$

$$\boxed{4s^2 + 160s \text{ m}^2}.$$

8. If a number plus its reciprocal equals $\frac{10}{3}$, find the number.

Solution:
Let the number be x.
Then,

$$x + \frac{1}{x} = \frac{10}{3}.$$

Multiplying by $3x$,

$$3x^2 + 3 = 10x.$$

Therefore,

$$3x^2 - 10x + 3 = 0.$$

Splitting the middle term,

$$3x^2 - 9x - x + 3 = 0$$

$$3x(x - 3) - 1(x - 3) = 0$$

$$(3x - 1)(x - 3) = 0.$$

Hence,

$$3x - 1 = 0 \quad \text{or} \quad x - 3 = 0.$$

Thus,

$$\boxed{x = \frac{1}{3} \text{ or } x = 3}.$$

9. A rectangular pool has area $2x^2 + 7x + 3$ square hastas. If its width is $2x + 1$ hastas, find its length.

Solution:

$$\text{Length} = \frac{\text{Area}}{\text{Width}}$$

$$= \frac{2x^2 + 7x + 3}{2x + 1}.$$

Factorising the numerator,

$$2x^2 + 7x + 3 = 2x^2 + 6x + x + 3$$

$$= 2x(x + 3) + (x + 3)$$

$$= (2x + 1)(x + 3).$$

Therefore,

$$\text{Length} = \frac{(2x + 1)(x + 3)}{2x + 1}$$

$$\boxed{x + 3 \text{ hastas}}.$$

10. If both $x - 2$ and $x - \dfrac{1}{2}$ are factors of $px^2 + 5x + r$, show that $p = r$.

Solution:
Since $x - 2$ and $x - \dfrac{1}{2}$ are factors,

$$px^2 + 5x + r = k(x - 2)\left(x - \frac{1}{2}\right).$$

Now,

$$(x - 2)\left(x - \frac{1}{2}\right) = x^2 - \frac{1}{2}x - 2x + 1$$

$$= x^2 - \frac{5}{2}x + 1.$$

Thus,

$$px^2 + 5x + r = k\left(x^2 - \frac{5}{2}x + 1\right).$$

Comparing the coefficients of x^2 and the constant terms,

$$p = k$$

and

$$r = k.$$

Hence,

$$\boxed{p = r}.$$

11. If $a+b+c=5$ and $ab+bc+ca=10$, prove that

$$a^3+b^3+c^3-3abc=-25.$$

Solution:
We know that

$$(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca).$$

Given,

$$a+b+c=5$$

and

$$ab+bc+ca=10.$$

Therefore,

$$25=a^2+b^2+c^2+20$$

$$a^2+b^2+c^2=5.$$

Hence,

$$a^2+b^2+c^2-ab-bc-ca=5-10=-5.$$

Using the identity

$$a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca),$$

we get

$$a^3+b^3+c^3-3abc=5(-5)$$

$$\boxed{a^3+b^3+c^3-3abc=-25}.$$

12. By factoring the expression, check that n^3-n is always divisible by 6 for all natural numbers n. Give reasons.

Solution:

$$n^3 - n = n(n^2 - 1)$$

$$= n(n-1)(n+1).$$

Thus,

$$n^3 - n$$

is the product of three consecutive integers:

$$(n-1), \quad n, \quad (n+1).$$

Among any three consecutive integers:

- one is divisible by 3;
- at least one is even and hence divisible by 2.

Therefore, their product is divisible by

$$2 \times 3 = 6.$$

Hence,

$$\boxed{n^3 - n \text{ is always divisible by } 6}.$$

13. Find the value of:

(i) $x^3 + y^3 - 12xy + 64$, when $x + y = -4$.

(ii) $x^3 - 8y^3 - 36xy - 216$, when $x = 2y + 6$.

Solution:

(i)

Given,

$$x + y = -4.$$

Using

$$(x + y)^3 = x^3 + y^3 + 3xy(x + y),$$

we get

$$(-4)^3 = x^3 + y^3 + 3xy(-4).$$

Therefore,

$$-64 = x^3 + y^3 - 12xy.$$

Hence,

$$x^3 + y^3 - 12xy + 64 = -64 + 64$$

$$\boxed{0}.$$

(ii)

Given,

$$x = 2y + 6.$$

Therefore,

$$x - 2y = 6.$$

Using

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b),$$

put $a = x$ and $b = 2y$:

$$(x - 2y)^3 = x^3 - 8y^3 - 6xy(x - 2y).$$

Since $x - 2y = 6$,

$$6^3 = x^3 - 8y^3 - 6xy(6).$$

Thus,

$$216 = x^3 - 8y^3 - 36xy.$$

Therefore,

$$x^3 - 8y^3 - 36xy - 216 = 0.$$

Hence,

$$\boxed{0}.$$

FINAL ANSWERS

Question	Answer
1	As simplified above
2	$357,\ 9984,\ 384,\ 3176523,\ 7880599,$ $2048383,\ -1225043,\ -26730899$
3	Factorised forms above
4	Simplified forms above
5	$(5a-3b)^2,\ (6s+7t)(6s-7t)$
6	$6(a+2b)(a-2b),\ 3p(s-1)(s-4)$
7	$4s^2+160s$
8	$3,\ \frac{1}{3}$
9	$x+3$ hastas
10	$p=r$
11	-25
12	Divisible by 6
13	$0,\ 0$